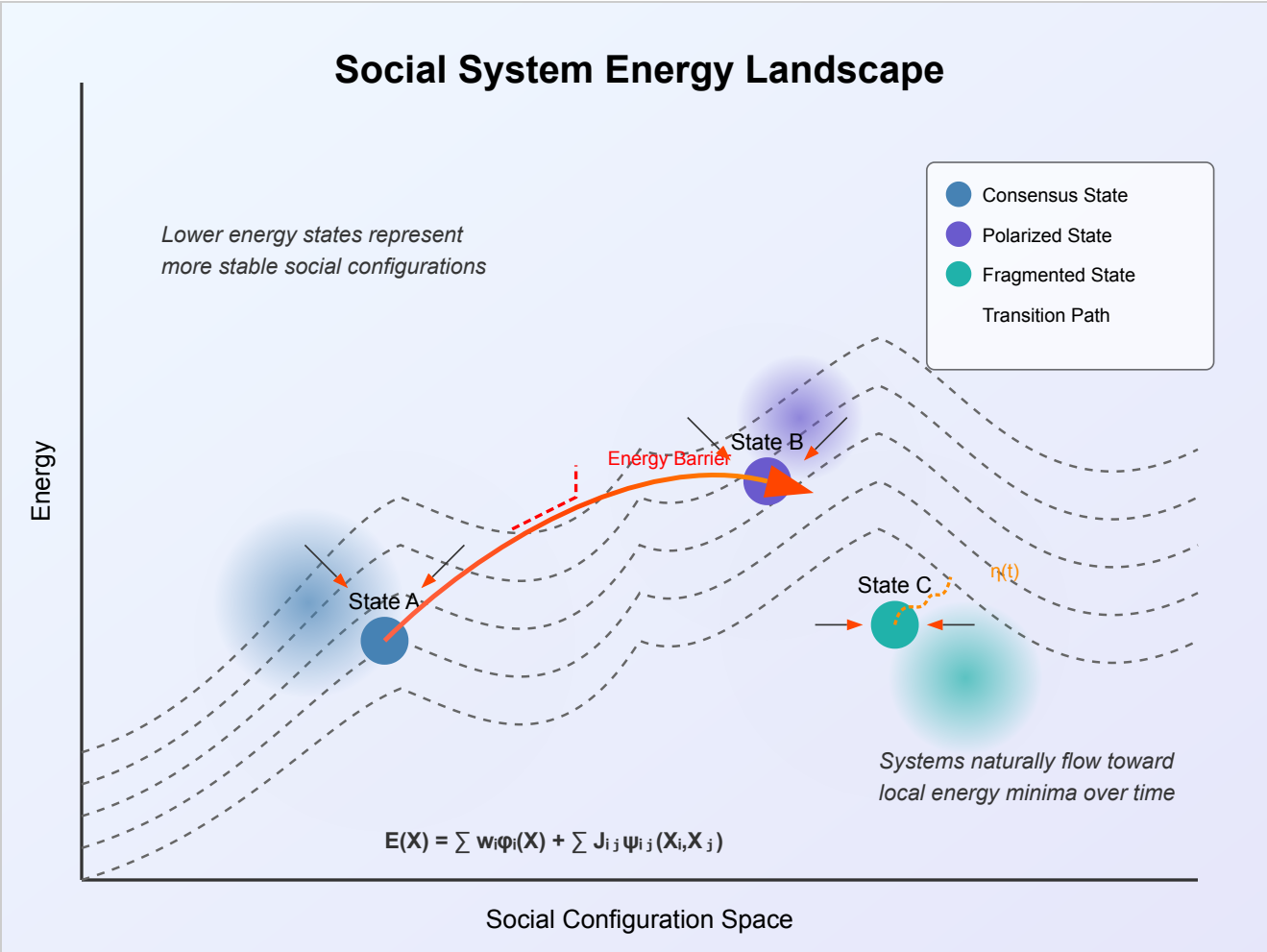

Drawing inspiration from physical systems, the framework conceptualizes social interactions as energy exchanges governed by conservation laws and gradient flows. This represents a fundamental shift from discrete, rule-based transitions to continuous energy landscapes where:

- Social configurations gravitate toward low-energy states
- Barriers between states create metastable social configurations
- Perturbations can push systems into new equilibrium conditions
- Phase transitions can model sudden social shifts

The energy function $E(X)$ defines a landscape over the state space, with social configurations naturally evolving toward locally minimal energy states according to the gradient flow:

$$\frac{dX}{dt} = -\nabla E(X).$$

This formulation allows for the modeling of complex social phenomena such as resistance to change, tipping points, and path-dependent evolution.

1.2.3 Temporal Resource Allocation